



Total Product Orders

Total Product Orders

20M
10M
0M



Raw Reorder Rate

Raw Reorder Rate

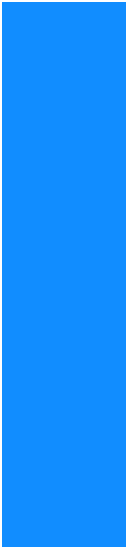
0.6
0.4
0.2
0.0



Low-Data Products

Low-Data Products

10K
5K
0K



Raw Reorder Rate by department

department

dairy eggs
beverages
produce
bakery
deli
pets
babies
bulk
snacks
alcohol
meat seafood
breakfast

0.0 0.2 0.4 0.6

Raw Reorder Rate

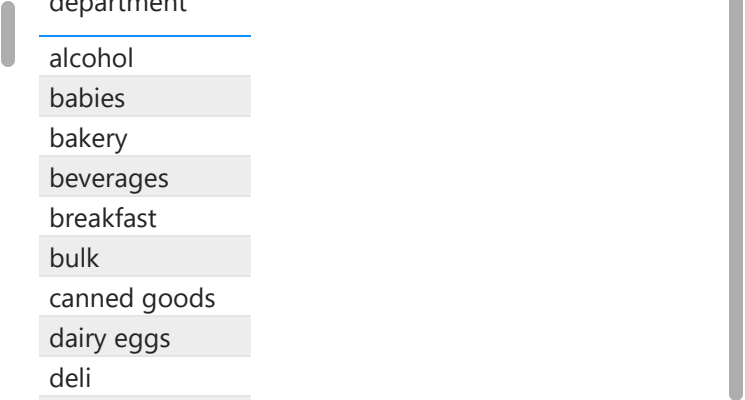


aisle

air fresheners candles
asian foods
baby accessories
baby bath body care
baby food formula
bakery desserts
baking ingredients
baking supplies decor
beauty
beers coolers
body lotions soap
bread
breakfast bakery

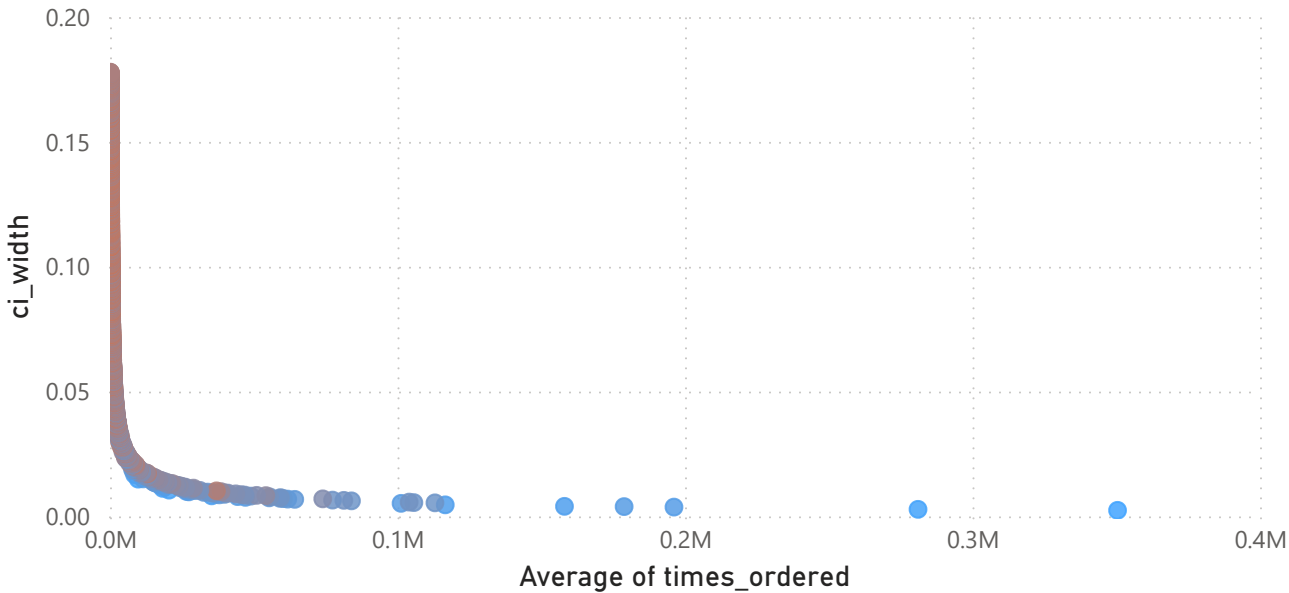
department

alcohol
babies
bakery
beverages
breakfast
bulk
canned goods
dairy eggs
deli
dry goods pasta
frozen
household
international
meat seafood

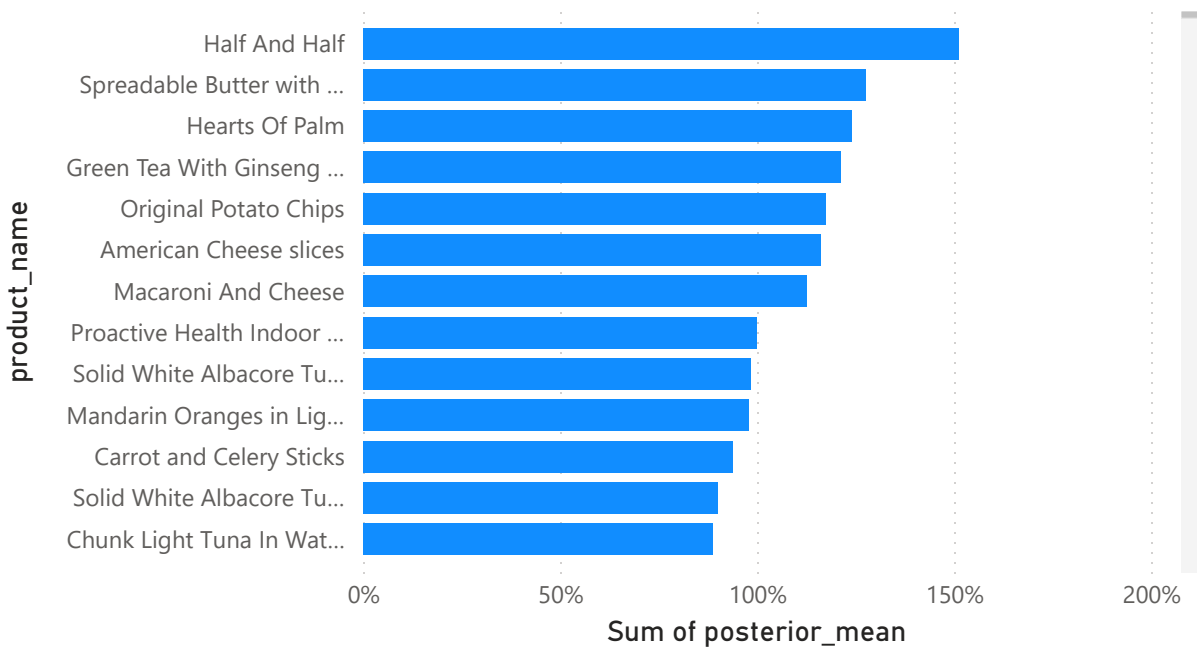


Average of times_ordered by ci_width

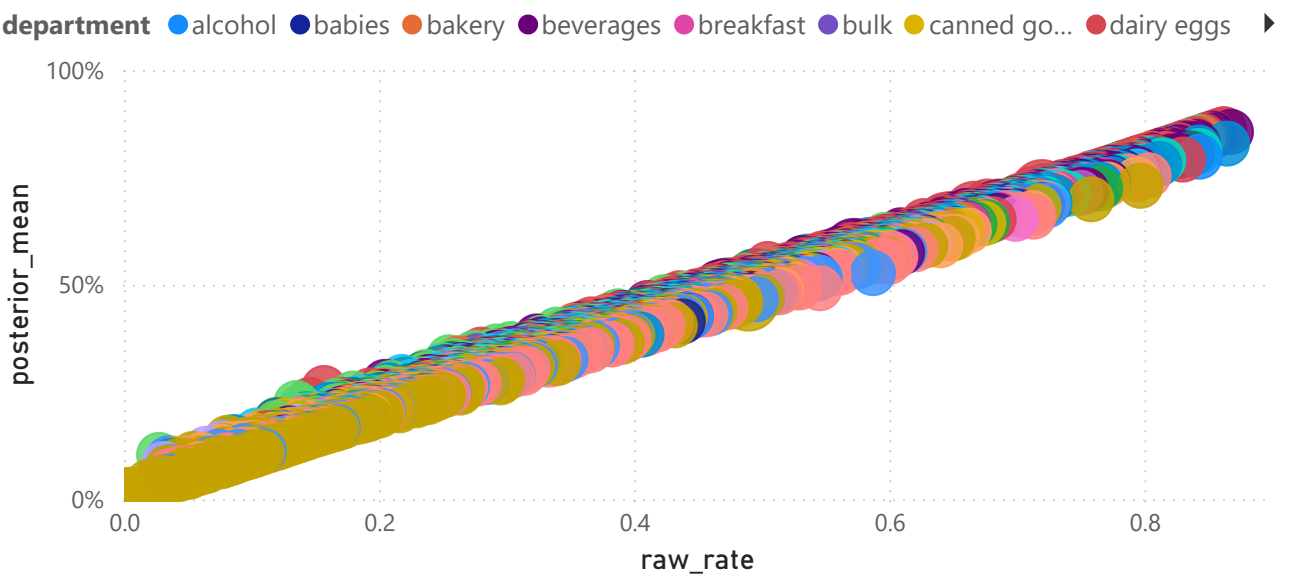
Average of posterior_mean 30.705% 84.340%



Sum of posterior_mean by product_name



Count of times_ordered by department, raw_rate and posterior_mean



Preparing Q&A

product_name	aisle	times_ordered	raw_rate	posterior_mean	ci_lower	ci_upper	ci_width
Half And Half Ultra Pasteurized	milk	2177	0.863	86.190%	0.847	0.876	0.029
Whole Organic Omega 3 Milk	milk	6796	0.862	86.160%	0.853	0.870	0.016
Organic Lactose Free Whole Milk	soy lactosefree	6289	0.859	85.830%	0.850	0.867	0.017
Fragrance Free Clay with Natural Odor Eliminator Cat Litter	cat food care	96	0.906	85.680%	0.788	0.914	0.126
Milk, Organic, Vitamin D	milk	14941	0.857	85.660%	0.851	0.862	0.011
Maca Buttercups	candy chocolate	81	0.926	85.610%	0.782	0.917	0.136
Lo-Carb Energy Drink	energy sports drinks	348	0.868	85.590%	0.818	0.890	0.072
Chocolate Love Bar	candy chocolate	73	0.932	85.450%	0.777	0.918	0.142
Organic Homogenized Whole Milk	milk	2950	0.854	85.310%	0.840	0.866	0.026
Ultra-Purified Water	water seltzer sparkling water	1101	0.854	85.160%	0.830	0.872	0.042
Organic Reduced Fat Milk	milk	26384	0.850	84.990%	0.846	0.854	0.009
Sparkling Water	water seltzer sparkling water	49	0.898	84.920%	0.756	0.923	0.166
Real2 Alkalized Water 500 ml	water seltzer sparkling water	324	0.855	84.770%	0.808	0.884	0.076
Organic Blueberry B Mega	refrigerated	76	0.895	84.670%	0.769	0.911	0.143

aisle

- ☐ air fresheners candles
- ☐ asian foods
- ☐ baby accessories
- ☐ baby bath body care
- ☐ baby food formula
- ☐ bakery desserts
- ☐ baking ingredients
- ☐ baking supplies decor
- ☐ beauty
- ☐ beers coolers

department

- ☐ alcohol
- ☐ babies
- ☐ bakery
- ☐ beverages
- ☐ breakfast
- ☐ bulk
- ☐ canned goods
- ☐ dairy eggs
- ☐ deli
- ☐ dry goods pasta
- ☐ frozen

Executive Summary: Bayesian Reorder Rate Analysis

Key Insights & Methodology

- **Raw Reorder Rate vs. Bayesian Estimation:** Standard reorder rates can produce extreme or biased estimates for products or categories with low order counts. Applying a Bayesian posterior mean provides a smoothed, risk-adjusted metric that prevents low-volume noise from skewing performance rankings.
- **Top Department Performance:** High-frequency categories such as **Dairy & Eggs**, **Beverages**, and **Produce** exhibit strong baseline reorder behavior, maintaining consistency across both raw and posterior estimates.
- **Low-Data Product Adjustment:** Products with limited order history are pulled back toward the global mean rate, ensuring marketing and inventory decisions are based on statistically reliable demand signals rather than random single-order spikes.

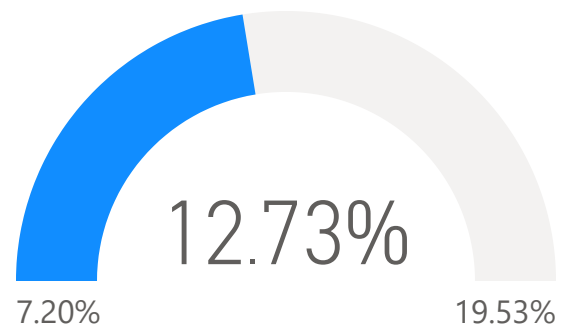
Strategic Recommendations

1. **Inventory Prioritization:** Focus fulfillment optimization and stock availability on top-performing reorder categories identified by the posterior mean.
2. **Targeted Promotions:** Use Bayesian scores to identify underperforming items with high true potential, optimizing promotional spend without over-investing in volatile low-volume products.

← Estimated Discrepancy Rate

12.73%

95% High-Density Interval (HDI) Confidence Range



MODERATE_MISCOUNT_RISK

First stock_risk_category

stock_risk_category	posterior_mean	hdi_95_lower	hdi_95_upper	Sum of posterior_alpha	Sum of posterior_beta
MODERATE_MISCOUNT_RISK	12.73%	7.20%	19.53%	14.00	96.00
Total				14.00	96.00